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Section: A Batch: 201 Date: _____**Experiment No. 08****Name of the Experiment: Determination of the Young's modulus of the given material bar by non-uniform bending using pin and microscope method.****Theory:**

Young's modulus is named after Thomas Young, 19th century, British scientist. In solid mechanics, Young's modulus is defined as the ratio of the longitudinal stress over longitudinal strain, in the range of elasticity the Hook's law holds (stress is directly proportional to strain). It is a measure of stiffness of elastic material.

If a wire of length L and area of cross-section ' a ' be stretched by a force F and if a change (increase) of length ' l ' is produced, then

$$\text{Young's modulus} = \frac{\text{Normal stress}}{\text{Longitudinal strain}} = \frac{F/a}{l/L}$$

Non Uniform Bending Using Pin and Microscope

Here the given beam (meter scale) is supported symmetrically on two knife edges and loaded at its centre. The maximum depression is produced at its center. Since the load is applied only one point of the beam, the bending is not uniform throughout the beam and the bending of the beam is called non-uniform bending.

In non-uniform bending (central loading), the Young's modulus of the material of the bar is given by

$$Y = \frac{mgl^3}{48I\theta} \quad (1)$$

I is the moment of inertia of the bar.

For a rectangular bar,

$$I = \frac{bd^3}{12} \quad (2)$$

Substituting (2) in (1), In non-uniform bending, the young's modulus of the material of the bar is given by,

$$Y = \frac{mgl^3}{4bd^3e} \quad (3)$$

Where m-Mass loaded for depression, g-Acceleration due to gravity, l-Length between knife edges, b-Breadth of the bar using Vernier calipers, d-Thickness of the bar using screw gauge, and e-Depression of the bar. Plot a graph e vs. l^3 (or e vs. m). From the graph l^3/e (or m/e) can be calculated.

Procedure for Simulation

1. Select the environment and material for doing experiment.
2. Choose mass, length, breadth and thickness of the material bar using sliders on the right side of the simulator.
3. Fix the distance between knife edges.
4. Focusing the microscope and adjusting the tip of the pin coincides with the point of intersection of the cross wires using left and top knobs on microscope respectively.
5. Readings are noted using the microscope reading for 0g. Zoomed part of microscope scale is available by clicking the centre part of the apparatus in the simulator. Total reading of microscope is $MSR + VSR \cdot LC$. MSR is the value of main scale reading of the microscope which is coinciding with the zero of Vernier scale. One of the division in the Vernier scale coincides exactly with the main scale is the value of VSR. LC is the least count.
6. Weights are added one by one say 50g, then pin moves downwards while viewing through microscope. Again adjust the pin such that it coincides exactly with the cross wire.
7. The readings are tabulated and Y is determined using equation (3).

Apparatus:

- Rectangular beam
- Microscope or travelling microscope
- Suitable weights
- Slide calipers
- Screw Gauge □ Two strong knife-edges □ Meter scale etc.

Experimental Data:

Number of divisions on the Vernier =

Vernier constant of the microscope =

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(A) Table for determining load versus depression data:

N o o f o b s.	Distanc e of the knife edges, l (cm)	Load , m (gm)	Readings of the microscope								Mean (cm.)	Depressi on for load, e (cm)	Mean e (cm)	l^3 (cm ³)	Mean l^3 (cm ³)
			Loading				Unloading								
			LSR x (cm)	V. S. R.	Value of V.S.R. y (cm)	Total, x+y (cm)	LSR x (cm)	V. S. R.	Value of V.S.R. y (cm)	Total, x+y (cm)					
1	10	50	4.95	25	0.025	4.975	4.95	25	0.025	4.975	X ₀ = 4.975	X ₄ -X ₀ = 0.003		1000	31625
2	15	100	4.95	15	0.015	4.965	4.95	30	0.03	4.98	X ₁ = 4.9725	X ₅ - X ₁ =0.0165	0.0238	3375	
3	20	150	4.95	20	0.02	4.97	4.95	20	0.02	4.97	X ₂ = 4.97	X ₆ -X ₂ = 0.035		8000	
4	25	200	4.95	23	.023	4.973	4.95	25	0.025	4.975	X ₃ = 4.974	X ₇ -X ₃ = 0.041		15625	
5	30	250	4.95	27	.027	4.977	4.95	30	0.03	4.98	X ₄ =4.978			27000	
6	35	300	4.95	40	.04	4.99	4.95	38	0.038	4.988	X ₅ = 4.989			42875	
7	40	350	4.95	50	.05	5.00	5	10	0.01	5.01	X ₆ = 5.005			64000	
8	45	400	5	15	.015	5.015	5	15	0.015	5.015	X ₇ = 5.015			91125	

Calculation:

Thickness of the material bar “d” = .45 cm

Breadth of the material bar “b” = 1.2 cm

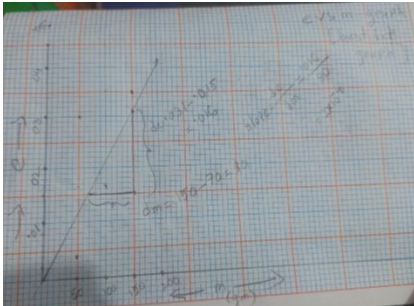
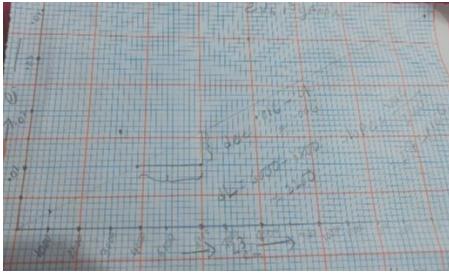
$$1328781.51$$

Mean value of l^3/e (from e vs l^3 graph) = 1328781.51 cm^2

Slope of e vs. m graph = $2 \times 10^{-4} \text{ cm/gm}$

Slope⁻¹ = $m/e = 5000 \text{ gm/cm}$

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$$\begin{aligned}
 Y &= \frac{e g}{4 b d^3} \times \frac{1}{d e} \times \frac{1}{d m^3} \\
 &= \frac{0.0238 \times 981}{4 \times 0.2 \times 4 \times 1.2 \times 953} \times 366666.66 \times 5000 \\
 &= 2.23 \times 10^{11} \text{ dyne/cm}^2 \\
 &= 2.23 \times 10^{10} \text{ Nm} \\
 \text{Difference} &= \left[2.23 \times 10^{10} - 2 \times 10^{10} / 2 \times 10^{10} \right] \times 100\% \\
 &= 88.85\% \\
 \therefore \text{Accuracy} &= 100 - 88.85\% \\
 &= 11.15\%
 \end{aligned}$$

Load applied for depression “e” = 0.0238 cm

Young’s modulus of the material bar,

$$Y = \frac{m g l^3}{4 b d^3 e}$$

$$= 3.54 \times 10^{-9} \text{ dynes/cm}^2 = \text{N/m}^2.$$

$$\text{Difference} = [(\text{Experimental Result} - \text{Theoretical Result}) / \text{Theoretical Result}] \times 100\% =$$

$$\text{Accuracy} = 100\% - \% \text{ Difference} = \mathbf{11.75\%}$$

Result:

The Young's modulus of the given material using non uniform bending method is, $Y = 2.23 \times 10^{10} \text{ N/m}^2$

Discussions:

Q: What is stress and strain? Define longitudinal stress and longitudinal strain. How are they related to Young's Modulus? Derive the unit of Young's Modulus.

Stress is the force that putted on to to object to calculate the force of the area .

Strain is the change in length divided by the original length of the area .Longitudinal force is the deforming force applied on to the body. Young's modules is equal to the longitudinal stress / strain .

$$Y = F_{lo}/A*(L_n - L_o).$$

The unit of Young's modules is N/m^2 .

Q: In determining the Young's Modulus of the material using the formula given which quantities you think should be measured with caution and why?

Quantities should be measured with cautions because it helps to get the value of load applied depression .

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Q: On what factors does the value of Y depend?

The value of Y depends upon temperature & pressure.

Q: Is it necessary to use a dead load here?

No, it is not necessary to use dead load here

Q: Suppose, you are provided with two wires both made of copper but they are of different length and diameter. What do you think about their Young's Modulus? Will they be different or same? Why?

Q: Why do the readings for load increasing and load decreasing differ?

It differs for component purpose

Q: Why is it necessary to wait a while before reading is taken after putting a load on the hanger?

It is necessary to wait because it will help the weight to elongate or contract fully

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Q: Which is more elastic, rubber or steel?

The steel has greater elasticity between rubber & Steel .

Q: What is the standard value of the *Young's Modulus* of the material used to perform the experiment?

The standard value is $2 \times 10^{11} \text{ N/m}^2$